

Prova Calculo 1 - S2

- 40 01. Determine a função derivada primeira $f'(x)$ e a função derivada segunda $f''(x)$ relativas a seguinte função:

$$f(x) = e^x \cdot (x - \text{Sen}x + \text{Cos}x).$$

- 30 02. Determine a equação reduzida da reta tangente t ao gráfico da função

$$y = f(x) = \frac{x^5 + 3}{x^3 - 2} \quad \text{no ponto de abscissa } x_0 = 1.$$

- 30 03. Determine a derivada implícita $\frac{dy}{dx}$ para os pontos $P(x, y)$ da curva φ que satisfazem a equação

$$x^3 + x^2 \cdot y^3 = y^5 + x^4.$$

01. $f(x) = e^x (x - \text{Sen}x + \text{Cos}x) \Rightarrow$

$\frac{df(x)}{dx} = e^x (x - \text{Sen}x + \text{Cos}x) + e^x (1 - \text{Cos}x - \text{Sen}x) \Rightarrow$

$\frac{df(x)}{dx} = e^x (x - \text{Sen}x + \text{Cos}x + 1 - \text{Cos}x - \text{Sen}x) \Rightarrow$

$\frac{df(x)}{dx} = e^x (x - 2\text{Sen}x + 1) = f'(x)$

$\frac{d^2f(x)}{dx^2} = e^x (x - 2\text{Sen}x + 1) + e^x (1 - 2\text{Cos}x) \Rightarrow$

$\frac{d^2f(x)}{dx^2} = e^x (x - 2\text{Sen}x + 1 + 1 - 2\text{Cos}x) \Rightarrow$

$\frac{d^2f(x)}{dx^2} = e^x (x - 2\text{Sen}x - 2\text{Cos}x + 2) = f''(x)$

$$2. f(x) = \frac{x^5 + 3}{x^3 - 2} \Rightarrow$$

$$f'(x) = \frac{(5x^4)(x^3 - 2) - (x^5 + 3)(3x^2)}{(x^3 - 2)^2} \Rightarrow$$

$$f'(x) = \frac{5x^7 - 10x^4 - 3x^7 - 9x^2}{(x^3 - 2)^2} \Rightarrow$$

$$f'(x) = \frac{2x^7 - 10x^4 - 9x^2}{(x^3 - 2)^2} = 7$$

$$f'(x) = \frac{x^2 (2x^5 - 10x^2 - 9)}{(x^3 - 2)^2}$$

$$f'(x_0) = m_t$$

$$f'(x_0) = f'(1) = \frac{1^2 (2 \cdot 1^5 - 10 \cdot 1^2 - 9)}{(1^3 - 2)^2} \Rightarrow$$

$$f'(x_0) = \frac{1 \cdot (2 - 19)}{(-1)^2} \Rightarrow$$

$$f'(x_0) = -17$$

$$x_0 = 1$$

$$f(x_0) = f(1) = \frac{1^5 + 3}{1^3 - 2} = \frac{4}{-1} = -4$$

$$y_0 = f(x_0) = -4$$

$$t: (y - y_0) = f'(x_0) \cdot (x - x_0)$$

$$t: (y + 4) = -17(x - 1)$$

$$t: y = -17x + 17 - 4$$

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03. $\varphi: x^3 + x^2 y^3 = y^5 + x^4 \Rightarrow$

$$\frac{d}{dx}(x^3 + x^2 y^3) = \frac{d}{dx}(y^5 + x^4) \Rightarrow$$

$$3x^2 + 2x \cdot y^3 + x^2 \cdot 3y^2 \frac{dy}{dx} = 5y^4 \frac{dy}{dx} + 4x^3 \Rightarrow$$

$$(3x^2 y^2 - 5y^4) \frac{dy}{dx} = 4x^3 - 3x^2 - 2xy^3 \Rightarrow$$

$$\frac{dy}{dx} = \frac{4x^3 - 3x^2 - 2xy^3}{3x^2 y^2 - 5y^4} \Rightarrow$$

$$\frac{dy}{dx} = \frac{x(4x^2 - 3x - 2y^3)}{y^2(3x^2 - 5y^2)}$$